

Orbit Wars — RL Math, v9

A complete, self-contained reference:

the mixed 2p/4p league with per-seat categorical ownership

2026-06-14

This document is the *standalone* specification of the v9 training program — the policy, the reward, the PPO + GAE update, the BC warm-start, and the mixed two-player / four-player Elo league. It supersedes the earlier delta-on-delta chain (`rl_math.tex` → `rl_math_v6.tex` → `rl_design_v7.tex` → the old `rl_math_v9.tex`) by inlining everything those documents established; **no back-references are required to read it**. Section 1 records the design history that produced each choice (small intermediate patches are omitted). Two mechanisms that the prior write-ups treated only in passing are given full treatment here: the **bilinear pointer head** (Section 5.1) and **why the WHERE distribution is a categorical and not a Dirichlet** — including how the KL trust region silently collapses, and how reward and loss blew up, in the earlier attempts (Section 7).

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1 Design history (what was tried, what broke, what stuck)

<i>Variable reference — this section</i>	
N	number of players, $N \in \{2, 4\}$ (symmetric Markov game)
GRPO	Group Relative Policy Optimization — the retired critic-free trainer (Section 9.4)
PPO + GAE	clipped actor-critic with Generalized Advantage Estimation (Section 9)
BC	behaviour-cloning warm-start (Section 10)
WHETHER / WHERE	the two action factors: per-source Bernoulli launch gate / categorical destination (Section 6)
PFSP	prioritized fictitious self-play matchmaking (Section 11)
FFA	four-player free-for-all scoring
KL	Kullback–Leibler divergence — the PPO trust-region measure
PopArt	output-preserving value normalisation (Section 9.1)

The v9 design is the fixed point of a long sequence of measured failures. Each item below records a decision and the failure that forced it; the math each one settled on is given in full in the body sections. Minor parameter patches are omitted.

(0) The game and the two-player special case. Orbit Wars is an N -player ($N \in \{2, 4\}$) symmetric Markov game (Section 2). Training began on the two-player zero-sum case and only generalised to mixed 2p/4p at v8/v9.

(1) GRPO from scratch — the passivity wall. The first trainer used **Group Relative Policy Optimization**: no critic, a group of trajectories from the same start/opponent, each return centred against its group mean (Section 9.4). With a *continuous* action actor (a squashed Gaussian over per-fleet direction/fraction controls) every from-scratch run collapsed into a **passive basin**: the policy stopped launching, beat only **random**, and the aiming gradient vanished. Re-shaping the reward only moved *which* basin it landed in.

(2) **PPO + GAE — escapes the GRPO wall, but the continuous actor blows up.** Adding a value head and Generalized Advantage Estimation (Section 9) broke the aiming freeze by giving per-step credit. But the continuous squashed-Gaussian actor was numerically fragile: summing its log-prob over $E \cdot 3K$ components, an extreme mean with a saturated sample over a tiny variance produced a huge log-density, an exploding gradient, and $\exp(\log \pi - \log \pi_{\text{old}}) \rightarrow \infty$ (Section 7, “earlier blow-ups”). A second failure mode was a *frozen* policy: the entropy bonus pinned σ at its cap, the sampled action stayed noise-dominated, the advantage decorrelated from the mean, and the policy-gradient terms cancelled ($\text{adv_std} = 1$ yet $\text{KL} \rightarrow 0$). Both needed log-std clamps, μ clamps inside the density, log-ratio clamps, advantage clamps, and a two-phase exploration schedule just to stay finite.

(3) **The target / allocation actor — aiming becomes a heuristic, over-launch becomes impossible.** The continuous direction-vector actor was retired in favour of a **per-source allocation actor**: each owned planet emits one row over destinations, aiming is a deterministic orbital-intercept heuristic (Section 3), and the action is structurally legal (a simplex row cannot over-dispatch the garrison). This removed the entire “learn legality / learn aiming from a penalty” problem.

(4) **v6 — gated allocation, potential shaping, PopArt, and the bilinear head.** The v6 notebook kept the trunk and simulator but changed four things that all carried forward to v9: (i) an explicit **Bernoulli launch gate** (WHETHER) on top of the allocation (WHERE), so “hold” is a first-class decision (Section 6); (ii) the four-channel *event* reward — which was farmable (capture-churn, dispatch-spam, hoarding milestones, all maximisable while losing) — replaced by **potential-based shaping** (Section 8); (iii) **PopArt** value normalisation, because the MSE critic target spanned a ± 1000 terminal spike against $O(10^{-2})$ per-step shaping (Section 9.1); and (iv) the destination head replaced by a **bilinear pointer** (Section 5.1). The data path was also moved fully on-GPU / bf16 to make the rollout simulator-bound.

(5) **v7 — the bootstrap was the wall, not the algorithm.** Every from-scratch run in the project’s history hit the same two walls, reproduced one last time as a controlled smoke: a **catastrophic first update** (iteration-1 KL 2.06, clip-fraction 0.98 — every sample clipped, launch rate annihilated $13.5 \rightarrow 1.0$ per step) and then a **passivity plateau** (gauntlet 0.000 for 30 iterations). The best from-scratch result ever obtained was a 0.47 win rate vs. **starter** after ~ 1400 iterations (~ 20 h). v7’s conclusion: *the optimisation machinery is fine; the starting point is fatal*. The fix is a **behaviour-cloning (BC) warm-start** (Section 10) that expresses the strongest scripted bot directly in the gated-allocation heads, plus three **PPO guardrails** (Section 9.2) that make the first updates structurally incapable of destroying it.

(6) **v7 — categorical WHERE (the decisive architecture change).** Warm-started PPO on the v6 *Dirichlet* WHERE head *destroyed* the BC policy (gauntlet $0.755 \rightarrow 0.104$) while the KL spiked to 2–9 *despite* a per-minibatch KL stop. The Dirichlet’s density near a simplex vertex is so steep that one optimiser step moves stored-action log-probs by whole nats — nothing bounds damage *per step*. Replacing the WHERE distribution with a **per-source categorical** (one-hot sampled destination) fixed it (gauntlet $0.76 \rightarrow 0.81$, Elo $938 \rightarrow 1099$, KL pinned in band). The full mechanism, the sensitivity probe, and the entropy/reward/loss blow-ups are Section 7.

(7) **v8 — the mixed 2p/4p league.** A checkpoint can dominate heads-up (1.000 gauntlet, 0.82 vs. the previous best) and still *lose* four-player free-for-all (29% top-rate vs. 58%), and the Kaggle leaderboard mixes both formats. v8 added N -player play (official seating and top-vs-second combat), a **dual correlated Elo**, a gap-driven match mix, invited anchored members, weak-point matchmaking, and a mixed 2p+4p best-checkpoint gauntlet — all legacy-compatible (observation unchanged, $F=101$; v7 weights drop in by collapsing every opponent to one enemy code).

(8) **v9 — per-seat categorical ownership.** v8’s collapsed ownership scalar cannot tell *which* opponent owns a body. v9 replaces it with a **4-channel ego-relative seat one-hot** ($F=104$) so the policy can learn per-enemy behaviour in FFAs (Section 4). Legacy ($F=101$) members are admitted through an observation adapter. This is the design documented here.

2 Problem setup

<i>Variable reference — this section</i>	
N	number of players, $\in \{2, 4\}$
$S^{(i)}$	player i ’s total ship count (planets + in-flight fleets); the terminal score
b	base index of the world’s 4-fold-symmetric quadrant group (seating)
$\text{owner}(\cdot)$	seat-slot $\rightarrow$ player-id map; $S_{\text{home}}=10$ is the starting home garrison
ego	the learner — always player 0; opponents occupy seats $1..N-1$
E	number of planet slots; a turn’s action factors over these
WHETHER / WHERE	per-source Bernoulli launch gate / categorical destination

Orbit Wars is an N -player symmetric Markov game, $N \in \{2, 4\}$. The deterministic engine seats N homes on one random group of a 4-fold-symmetric quadrant arrangement, and every turn all players *simultaneously* issue fleet *launches*; the engine then moves fleets, resolves collisions/combat, rotates planets, produces ships, and (on a fixed schedule) spawns comets. Scoring is a **free-for-all**: at termination player i ’s score is its total ship count $S^{(i)}$ (planets + in-flight fleets). In 2p this is the zero-sum margin $S^{(0)} - S^{(1)}$; in 4p the objective is *ordinal* (be the maximum of four), which is not zero-sum and motivates the placement-linear outcome of Section 8.

Seating. The official engine records the world’s symmetric quadrant group’s base index b ; `reset(worlds, n_players)` seats owners exactly like the engine:

$$N=2 : \text{owner}(b) = 0, \text{owner}(b+3) = 1; \quad N=4 : \text{owner}(b+j) = j, j = 0..3; \quad S_{\text{home}} = 10.$$

Under the board’s 4-fold rotational symmetry every seat is fair in both formats. The learner is always player 0 (the ego); opponents occupy seats $1..N-1$.

The turn as a factored action. A turn’s action factors over the E planet slots: per *owned* source planet s the policy decides (i) **WHETHER** to launch (a Bernoulli gate) and (ii) **WHERE** to send (a categorical over destination planets). On a launch the *full* garrison is dispatched to the chosen destination, aimed by a deterministic heuristic. Unowned/empty sources contribute nothing. There is no continuous fraction, no per-slot launch budget, and no separate aiming output — the design history above retired all three.

3 N -player environment and the GPU simulator

Variable reference — this section

$r_\odot$	sun-disk radius, <code>SUN_RADIUS</code> = 10; board is 100×100
ν (also n)	ship count of a fleet
$v(\nu), v_{\max}$	fleet speed law and cap ($v_{\max}=6$), Eq. (1)
r_d, ϕ_0, w, C	destination orbit: radius, phase, angular velocity, board centre
$P(t)$	destination position at flight time t
$c_{\text{src}}, r_{\text{src}}$	source planet centre and radius
$t^*, g(t)$	intercept time (earliest root) and the intercept residual
θ	launch heading, atan2 of the lead point
$A_e^{(p)}$	ships of player p arriving at planet e
surv_e	survivors at e = top-minus-second arrival (official resolve)
E_c	<code>planet_cap</code> — fixed Structure-of-Arrays slot count
B	batch size: number of parallel worlds rolled out on-GPU

The board is 100×100 with a fixed central sun disk of radius $r_\odot = \text{SUN_RADIUS} = 10$. Planets orbit the centre at a fixed angular velocity; fleets fly straight at a ship-count-dependent speed; production adds ships each tick; comets spawn on a fixed schedule. All quantities are deterministic.

Fleet speed law. A fleet of ν ships moves at

$$v(\nu) = \min\left(v_{\max}, 1 + (v_{\max} - 1)\left(\frac{\ln \max(1, \nu)}{\ln 1000}\right)^{1.5}\right), \quad v_{\max} = 6. \quad (1)$$

Lead-target aiming (deterministic orbital intercept). A launch $s \rightarrow d$ aims not at d 's current position but at where the orbiting destination will be on arrival. With d at orbital radius r_d , phase ϕ_0 , angular velocity w about centre C , its position at flight time t (ticks) is $P(t) = C + r_d(\cos(\phi_0 + wt), \sin(\phi_0 + wt))$. A fleet leaves the source surface (a point at $r_{\text{src}} + 0.1$ from the source centre along the heading) and flies at $v = v(n)$, so the intercept time t^* is the earliest root of

$$g(t) = \|P(t) - c_{\text{src}}\| - (r_{\text{src}} + 0.1) - vt = 0, \quad (2)$$

solved by bracketed bisection (the earlier fixed-point iterate was not a contraction for fast-orbiting targets and missed ~20% of reachable orbiting planets). The heading is $\theta = \text{atan2}(P(t^*) - c_{\text{src}})$. The static-target case ($w=0$) recovers the straight-line aim.

Combat: official top-vs-second, N players. Arrivals at planet e are summed per player, $A_e^{(p)} = \sum_{\text{fleets} \rightarrow e, \text{owner}=p} n$, and the official resolve is the top-two statistic

$$\text{surv}_e = A_e^{(1\text{st})} - A_e^{(2\text{nd})}, \quad \text{owner}_e^{\text{surv}} = \begin{cases} \arg \max_p A_e^{(p)} & \text{surv}_e > 0 \\ \emptyset & \text{tie,} \end{cases} \quad (3)$$

implemented as a (B, E, N) scatter-add + `topk(2)`; survivors reinforce or attack the garrison. For $N=2$ this is *algebraically identical* to the pairwise max/min resolve.

Batched GPU simulator. The whole game runs as batched LibTorch tensor ops on the GPU (`GpuEnv`): encode, sample, decode, step, and scripted opponents all stay on-device, so the loop is GPU-compute-bound (no host round-trips). Planets live in $E_c = \text{planet_cap}$ fixed Structure-of-Arrays slots (last few reserved for the single live comet group); fleets in a fixed pool with an alive

mask. Because the per-planet actor is permutation-equivariant, no entity sort is needed. Integer quantities are held in `float32` (exact below 2^{24}).

Parity. `GpuEnv` is bit-exact to the official Kaggle engine including official comets: world-generation is exact, full games are tick-exact, and scripted-launch decisions match to 99.8% (the residual being exact-tie collision flips, a benign `float32` boundary choice). The threat encoder and reachability mask use the *same* ray/disk projection in rollout and update, so the natively trained policy behaves identically when served from pure Python.

Seat interface and rollout boundary. `env_step(env, a, seats=[...])` takes, per opponent pid $1..N-1$, either a scripted code (vectorised on-GPU, zero forwards) or a frozen neural member acting per-step under `no_grad` via its own `env_encode(env, pid)` (the same encode, ego = that seat; the observation of Section 4 is seat-symmetric by construction). Termination is the step cap, ≤ 1 player alive, or ego death.

4 Observation encoding

<i>Variable reference — this section</i>	
$X \in \mathbb{R}^{E \times F}$	per-planet entity matrix; E planet slots, $F=104$ body features
g	(i) ego seat index; (ii) board-globals vector $\in \mathbb{R}^G$, $G=10$ (disambiguated by context)
m, u	entity mask ($m_e=1$ real body) / action mask ($u_e=1$ ego-owned with > 0 ships)
S, ϱ, d_c, ω	garrison, planet radius, distance-to-centre, board angular velocity
$\ell(n)$	shared ship-log $\ln(1+n)/\ln 1000$
$\mathbf{b}[e]$	ego-relative seat one-hot, Eq. (4)
o_e, ρ_e	owner of body e ; ego-relative seat index $\rho_e=(o_e-g) \bmod N$
N	<code>n_players</code>
$t_{ca}, p^2, t_{hit}, \tau$	threat ray/disk projection: closest-approach time, perp-distance ² , hit time, ETA
$\text{sgn}, \tilde{\nu}$	threat-slot friend/foe sign and log-size $\ell(\nu)$
N_SOON, N_BIG	ranked threat-list sizes (soonest / largest), 15 each
b^{legacy}	collapsed scalar owner code $\in \{0, 1, 2\}$ for the legacy obs adapter

For ego player g and state s the encoder emits four arrays: the per-planet entity matrix $X \in \mathbb{R}^{E \times F}$ (E planet slots, $F = 104$ features); the board globals $g \in \mathbb{R}^G$ ($G = 10$, a *separate* input, not replicated into rows); the entity mask $m \in \{0, 1\}^E$ ($m_e = 1$ iff slot e holds a real body); and the action mask $u \in \{0, 1\}^E$ ($u_e = 1$ iff e is ego-owned with > 0 ships). m hides padding inside attention; u selects the owned sources that act and is also a body feature. Each row is 14 body features (v9: $11 \rightarrow 14$ from the ownership change) followed by a 90-feature incoming-threat block.

Body features (per planet). Each value is normalised into roughly $[-1, 1]$. With S the garrison, ϱ the radius, d_c the distance to centre, ω the board angular velocity, and $\ell(n) = \ln(1+n)/\ln 1000$ the shared ship-log:

#	feature	encoded value	meaning
0	position x	$p_x/100$	horizontal position, $[0, 1]$
1	position y	$p_y/100$	vertical position, $[0, 1]$
2	orbital speed	$ \omega d_c/6$ (0 if static)	tangential orbit speed vs. $v_{\max}=6$
3	spin sign	+1 ccw / -1 cw / 0	rotation direction
4	radius	$\varrho/3$	capture / collision disk size
5	production	prod/5	ships made per tick
6	ships	$\ell(S)$	garrison, log-compressed
7–10	seat one-hot	Eq. (4)	ego-relative owner (v9 change)
11	dist. to centre	$d_c/70.71$	radial distance from the sun
12	comet flag	$\mathbb{K}[\text{comet}]$	1 for a comet, else 0
13	action mask u_e	$\mathbb{K}[\text{ego} \wedge S > 0]$	ego-owned with ships (the decode key)

Radius and production are both fed (redundant for a normal planet, $\varrho = 1 + \ln \text{prod}$) so the net need not invert the log; the comet flag is the only observable comet signal (the engine hides comet lifetime), and it marks where the radius law breaks (a comet carries fixed COMET_PRODUCTION= 1 at an unrelated radius).

Per-seat categorical ownership (the v9 change). A single scalar code $\{0, 1, 2\}$ (neutral/ego/enemy) cannot say *which* enemy owns a body, and the ego-relative formula does not extend: in 4p, $(o - g) \bmod 2$ *aliases* seat $g+2$ onto the ego code. v8 collapsed every non-ego owner to one code 2 (the deployed everyone-vs-me view, $F=101$). v9 replaces the scalar with a 4-channel ego-relative seat one-hot. With $N = \mathbf{n_players}$, ego seat g , and per-body owner o_e , the ego-relative seat index $\rho_e = (o_e - g) \bmod N$ and

$$\mathbf{b}[e] = \left(\mathbf{1}[o_e \geq 0 \wedge \rho_e = 0], \mathbf{1}[o_e \geq 0 \wedge \rho_e = 1], \mathbf{1}[o_e \geq 0 \wedge \rho_e = 2], \mathbf{1}[o_e \geq 0 \wedge \rho_e = 3] \right), \quad (4)$$

i.e. $[\text{self} \mid \text{enemy seat}+1 \mid +2 \mid +3]$ with neutral = all-zero, so $F = 101 \rightarrow 104$. Properties:

- **categorical, not ordinal:** a scalar $2..n$ code imposes a fake magnitude/ordering across seats; one-hot channels are permutation-symmetric inputs the network can attach independent behaviour to (per-enemy threat assessment, target selection);
- **rotation-equivariant:** the same *relative* seat always lands in the same channel from any ego seat;
- **format-generalising:** in 2p only the first two channels ever fire, so 4p input space is a strict superset; when a seat is eliminated mid-FFA its channel stops firing, so 3-player intermediate states stay well-formed without being trained explicitly;
- **no aliasing:** ρ is mod N , so no opponent collides with the ego channel.

Everything else is owner-pooled: the threat block keeps the *signed* friend/foe convention (+1 ego, -1 any opponent), and the globals pool *all* opponents. Seat identity enters only through Eq. (4).

Board-summary globals $g \in \mathbb{R}^{10}$. One vector summarising the board, embedded once and broadcast to the heads (not copied into rows): game phase step/ T ; board angular velocity; ego total garrison $\ell(\cdot)$; enemy total garrison $\ell(\cdot)$ (pooled over all opponents); ego / enemy / neutral planet shares; ego / enemy in-flight ship totals $\ell(\cdot)$; and planet-slot occupancy $\min(n_p, E)/E$.

Incoming-fleet threat block (90 features/row). A purely reactive policy cannot defend what it cannot see, so each planet’s row carries a compact list of *its own* inbound fleets. Every in-flight fleet is projected to the planet it will hit and when, via the ray/disk test: for a fleet at o with unit heading $\hat{d}$ and a planet at c with radius ϱ , $t_{\text{ca}} = -\langle o - c, \hat{d} \rangle$, $p^2 = \|o - c\|^2 - t_{\text{ca}}^2$, $t_{\text{hit}} = t_{\text{ca}} - \sqrt{\varrho^2 - p^2}$

(if $t_{ca} \geq 0$, $p^2 \leq \varrho^2$), ETA $\tau = t_{hit}/v(\nu)$, dropped if the ray first crosses the sun. Among the fleets assigned to a planet we keep two ranked lists — the `N_SOON`= 15 soonest and the `N_BIG`= 15 largest — and each of the 30 slots contributes $[\text{sgn}, \tau/100, \tilde{v}]$ with $\text{sgn} \in \{+1 \text{ mine}, -1 \text{ enemy}, 0 \text{ empty}\}$ and $\tilde{v} = \ell(\nu)$, for $30 \times 3 = 90$ features. Empty slots are all-zero ($\text{sgn}=0$ already marks “no fleet”). The encoder itself has no attention; cross-planet reasoning is added by the policy trunk.

Legacy members: the obs adapter. v9’s league still admits v7/v8-format checkpoints ($F=101$). They are wrapped in `LegacyObsAdapter`, which collapses the one-hot back to the scalar they were trained on before their forward:

$$b^{\text{legacy}} = \mathbf{b}[0] + 2 \min(1, \sum_{k=1}^3 \mathbf{b}[k]) \in \{0, 1, 2\}, \quad (5)$$

exact on every reachable state (the channels are disjoint indicators). The wrapper is unwrapped on league-state save and re-wrapped on load. The *learner* cannot warm-start from $F=101$ weights (F changed), so v9 runs its own BC warm-start (the BC expert is scripted, so nothing else changes).

5 Policy network

<i>Variable reference — this section</i>	
d	trunk hidden width, <code>HIDDEN</code> = 512
$h_e^{(0)}$	projected token, $W_{\text{proj}}x_e + b_{\text{proj}}$
$\tilde{h}$	pre-LayerNormed tokens fed to attention
W_Q, W_K, W_V, W_O	self-attention query/key/value/output projections ($d \rightarrow d$)
$A, \mathcal{M}_{\text{ent}}$	attention output; additive key-mask ($-\infty$ on padding planets)
$\mathcal{B}(z)$	residual GLU block (trunk variant A)
$\mathcal{M}(z)$	pre-LN residual MLP block; n_{res} such blocks
$g' \in \mathbb{R}^{d_g}$	embedded globals, $d_g=32$; head input $h'_e=[h_e; g']$
$\ell_{s,d}$	bilinear WHERE logit, Eq. (7) (head $W_Q, W_K \in \mathbb{R}^{d \times (d+d_g)}$)
η_s	per-source gate (WHETHER) logit, bias <code>INIT_GATE_BIAS</code> = -1
$\tilde{V}_\theta(s), V(s)$	normalised value-head output; usable value $V=\sigma \tilde{V}_\theta + \mu$ (Section 9.1)
n_v	<code>VALUE.RES.BLOCKS</code> in the critic

The network maps the per-planet entity matrix $X \in \mathbb{R}^{E \times F}$ to per-planet action parameters; the G globals enter through a small separate path. The **trunk** is **shared** by the actor heads and the value head; only the heads differ.

Trunk. In order: input projection $h_e^{(0)} = W_{\text{proj}}x_e + b_{\text{proj}}$; one **masked cross-planet self-attention** block; an optional residual GLU block (trunk variant A); a stack of n_{res} pre-LayerNorm residual MLP blocks; a final LayerNorm. The projection and the MLP/GLU blocks are per-row (weights shared across planets); attention is the only cross-row mixing.

(1) Masked self-attention. On pre-LayerNormed tokens $\tilde{h} = \text{LN}(h^{(0)})$,

$$A = \text{softmax}\left(\frac{(\tilde{h}W_Q)(\tilde{h}W_K)^\top}{\sqrt{d}} + \mathcal{M}_{\text{ent}}\right)(\tilde{h}W_V), \quad h^{(1)} = h^{(0)} + A W_O, \quad (6)$$

with the key-mask $\mathcal{M}_{\text{ent}}[\cdot, j] = -\infty$ for padding planets ($m_j=0$). This lets a planet read the others’ positions/ownership/garrisons/threats — cross-planet board reasoning. It is computed with a fused flash kernel under bf16 autocast (the dead-planet mask supplied as a -10^9 additive bias). An **ARCH**

switch selects either this single-attention “trunk” or a true multi-head “transformer” stack of pre-LN attention + MLP layers; both feed identical heads.

(2) Optional residual GLU (variant A): for $z \in \mathbb{R}^d$, $\mathcal{B}(z) = z + W_o((W_v z + b_v) \odot \sigma(W_g z + b_g)) + b_o$. The sigmoid gate passes, attenuates, or suppresses each feature channel conditionally (e.g. amplify the threat channels when a planet is under attack); the residual keeps optimisation easy. Variant B replaces it with a plain residual MLP block.

(3) Pre-LayerNorm residual MLP blocks. $\mathcal{M}(z) = z + W_b \text{relu}(W_a \text{LN}(z)) + b_b$, then a final LayerNorm. The pre-norm is load-bearing: without it the deep trunk drove the head pre-activations large, the gate sigmoid saturated, and gradients vanished — pre-LN keeps activations bounded so the heads stay responsive.

Globals path. $g' = \text{relu}(W_g g + b_g) \in \mathbb{R}^{d_g}$, $d_g = 32$, broadcast to every planet and concatenated at the heads as $h'_e = [h_e; g'] \in \mathbb{R}^{d+d_g}$, so the trunk stays purely per-planet.

5.1 The bilinear (pointer) WHERE head

The WHERE head produces, for each owned source s , a logit $\ell_{s,d}$ for sending to each candidate destination d . The naive design (the original target actor) was a single **Linear**($d+d_g \rightarrow E$): $\ell_{s,d} = w_d^\top h'_s + b_d$, i.e. a *fixed weight vector w_d per destination slot index*. That is the wrong inductive bias here. Planet slots are a permutation-equivariant Structure-of-Arrays (Section 3) — the slot *index* carries no meaning, so a per-slot weight w_d has nothing stable to attach to; the head would have to memorise geometry it is not even shown, and is not equivariant under relabelling destinations.

v6 replaced it (flag `DEST_HEAD="bilinear"`) with a **pairwise pointer score** computed from *both* planets’ tokens. With a query projection W_Q and a key projection W_K , both $\mathbb{R}^{d \times (d+d_g)}$,

$$\ell_{s,d} = \frac{(W_Q h'_s)^\top (W_K h'_d)}{\sqrt{d}} \quad h'_e = [h_e; g']. \quad (7)$$

This is exactly a single attention score, but *used as the WHERE logits* over destinations rather than to mix features (the source is the query, every planet is a key). Why this is the right head:

- **Relational, content-based.** The score for $s \rightarrow d$ depends on *both* planets’ features — relative geometry (from positions/distance-to-centre), ownership (the seat one-hot of Eq. (4)), garrison, and inbound threat — which is precisely the information that decides “is d a good target for s ”. A fixed per-slot vector cannot express “send to the nearest weakly-held enemy planet”; a bilinear source $\times$ destination score can.
- **Permutation-equivariant in destinations.** Relabelling planets permutes the keys $\{W_K h'_d\}$ and therefore permutes the logits $\{\ell_{s,d}\}$ consistently — the head respects the SoA symmetry the linear head broke.
- **Shares the aiming mechanism.** It is the same query/key bilinear form as the trunk’s cross-planet attention, so “which planet to point at” is computed by content match, the operation the trunk attention was added to enable.
- **Calm init.** W_Q, W_K are scaled by `INIT_DEST_SCALE`= 0.02 so $\ell \approx 0$ (near-uniform WHERE) at start; “hold by default” is enforced by the gate bias below (Section 6), not by a diagonal logit, so the WHERE head starts unopinionated and learns to commit.

These logits are masked (alive $\wedge$ sun-reachable $\wedge$ no-self) and fed to the per-source *categorical* of Section 6. The bilinear head is identical for the (retired) Dirichlet WHERE and the (current) categorical WHERE — only the distribution placed on $\ell_{s,\cdot}$ changed (Section 7).

5.2 Gate head and value head

Gate head (WHETHER). `gate_head`, a `Linear( $d+d_g \rightarrow 1$ )`, emits a per-source fire logit η_s , with bias initialised to `INIT_GATE_BIAS` = -1 so few planets fire at start (Section 6).

Value head (the PPO critic). A masked-mean pool of the post-trunk tokens over live planets, concatenated with g' , then an input projection $d+d_g \rightarrow d$, `VALUE_RES_BLOCKS` pre-LN residual MLP blocks (the same block as the trunk), and a $d \rightarrow 1$ readout. Its scalar output is the *normalised* value $\tilde{V}_\theta(s)$; the usable value is $V(s) = \sigma \tilde{V}_\theta(s) + \mu$ with the PopArt statistics of Section 9.1.

Layer inventory (variant A).

module	layers (name: shape)	count
input projection	<code>proj</code> : $F \rightarrow d$	1
self-attention	<code>attn_{q,k,v,o}</code> : $d \rightarrow d$	4
LayerNorm (pre-attn, per-res, final)	<code>ln_attn</code> , <code>ln_resi</code> , <code>ln_out</code> : d	$n_{\text{res}}+2$
residual GLU (variant A)	<code>glu_{gate,val,out}</code> : $d \rightarrow d$	3
residual MLP $\times n_{\text{res}}$	<code>resi_{a,b}</code> : $d \rightarrow d$	$2n_{\text{res}}$
globals embed	<code>g_embed</code> : $G \rightarrow d_g$	1
bilinear WHERE	<code>dest_q</code> , <code>dest_k</code> : $(d+d_g) \rightarrow d$, Eq. (7)	2
gate head	<code>gate_head</code> : $(d+d_g) \rightarrow 1$	1
value in-proj / res / readout	<code>val_in</code> , <code>val_res_{a,b}</code> , <code>val_out</code>	$3n_v+2$

The deployed scale is $d=512$, $n_{\text{res}}=6$ (or the transformer $d=512$, $L=4/5$). The served twin must mirror trunk, globals path, and heads bit-for-bit; it serves the actor only (no critic at inference). Inference is compute-bound and far under the Kaggle 1 s/move budget at every scale up to $\sim 100\text{M}$ params (Section 12).

6 Action distribution: gated categorical allocation

<i>Variable reference — this section</i>	
$R_{s,d}$	reachability mask : $\mathbb{K}[\text{straight shot } s \rightarrow d \text{ clears the sun}]$ (sun-occlusion test below)
$c_\odot, r_\odot$	sun-disk centre and radius ($r_\odot=10$)
$\hat{u}, t, p^2$	ray parameters of the $s \rightarrow d$ occlusion test
$\ell_{s,d}, \tilde{\ell}_{s,d}$	bilinear WHERE logit / its masked form ($-\infty$ off-support)
d_s	sampled destination (categorical); the action row is its one-hot
m_d	alive mask of destination d
η_s, p_s, f_s	gate logit / fire probability $\sigma(\eta_s)$ / Bernoulli fire draw
ε	<code>GATE_EPS</code> = 0.02 (train-time gate clamp)
$\mathcal{H}_{\text{Bern}}, \mathcal{H}_{\text{cat}}$	gate Bernoulli / WHERE categorical entropies
n_{s,d_s}, S_s	ships dispatched / source garrison; floor <code>MIN_LAUNCH_SHIPS</code>

Per owned source s the policy is a product of two factors — a categorical **WHERE** over destinations and a Bernoulli **WHETHER** (fire) gate — and the policy factorises over owned sources. One structural mask encodes the sun-occlusion heuristic the model would otherwise relearn.

Reachability mask $R_{s,d}$ (sun occlusion). A straight shot $s \rightarrow d$ is absorbed by the sun (disk centre $c_\odot$, fixed radius $r_\odot=10$) iff the segment crosses that disk before reaching d . With $\hat{u} = (p_d - p_s) / \|p_d - p_s\|$, $t = \langle c_\odot - p_s, \hat{u} \rangle$, $p^2 = \|c_\odot - p_s\|^2 - t^2$, the shot is blocked iff $t \geq 0$, $p^2 \leq r_\odot^2$, and $0 \leq t - \sqrt{r_\odot^2 - p^2} < \|p_d - p_s\|$. Then $R_{s,d} = \mathbb{K}[\text{not blocked}]$. R is recomputed from the encoded positions, so it is identical in rollout and update with no extra buffers.

WHERE: per-source categorical. The bilinear logits $\ell_{s,d}$ of Eq. (7) are masked to alive, sun-reachable, non-self destinations and a categorical is placed on them:

$$\tilde{\ell}_{s,d} = \begin{cases} \ell_{s,d} & d \neq s \wedge m_d \wedge R_{s,d} \\ -\infty & \text{else,} \end{cases} \quad d_s \sim \text{Categorical}(\text{softmax}(\tilde{\ell}_{s,\cdot})), \quad (8)$$

and the sampled action row is the *one-hot* of d_s . (The masking forces zero probability on unreachable/padding/self destinations.)

WHETHER: Bernoulli gate. With the gate logit η_s ,

$$\text{train: } p_s = \text{clip}(\sigma(\eta_s), \varepsilon, 1-\varepsilon); \quad \text{deploy: } p_s = \text{clip}\left(\frac{\sigma(\eta_s)-0.2}{0.6}, 0, 1\right); \quad f_s \sim \text{Bernoulli}(p_s), \quad (9)$$

with $\varepsilon = \text{GATE_EPS} = 0.02$. Deploy uses a dead-zone / saturation trim so the gate can express “essentially never/always fire”; **training must not** — applying the hard trim at train time gives clip zero gradient outside the band, which starves both the gate head and (through the $f_s \cdot \log \pi_{\text{where}}$ coupling below) the WHERE head as gate logits sharpen. The smooth training form is the verified fix (gradient norm stayed in $[0.7, 5.4]$ over a 30-iteration smoke where the pre-fix run decayed monotonically toward zero). The greedy boundary $p_s \geq 0.5 \Leftrightarrow \eta_s \geq 0$ is identical in both.

Log-prob and entropy (WHERE counts only on a fire).

$$\log \pi_\theta(a \mid s) = \sum_{s: \text{owned}} \left[\underbrace{f_s \log p_s + (1-f_s) \log(1-p_s)}_{\text{gate}} + f_s \log \text{softmax}(\tilde{\ell}_{s,\cdot})_{d_s} \right], \quad (10)$$

$$\mathcal{H}[\pi_\theta(\cdot \mid s)] = \sum_{s: \text{owned}} \left[\mathcal{H}_{\text{Bern}}(p_s) + p_s \mathcal{H}_{\text{cat}}(\text{softmax}(\tilde{\ell}_{s,\cdot})) \right]. \quad (11)$$

On a hold ($f_s=0$) the destination is unobserved, so it must not be scored — otherwise a holding planet would be credited/penalised for a launch it never made. The WHERE entropy is symmetrically weighted by p_s .

Decode. A firing source dispatches its **full garrison** to the single drawn destination (the categorical row is one-hot), aimed by the orbital intercept of Section 3: $n_{s,d_s} = S_s$ ships if $f_s=1$ and $n_{s,d_s} \geq \text{MIN_LAUNCH_SHIPS}$, else held. A holding source keeps everything. Because the destination is a real, sun-reachable planet, over-dispatch and phantom launches cannot arise — the action is structurally legal. Multi-target pressure comes over *time*: the gate is a per-turn fire *rate*, the garrison refills by production, and the source can fire again next turn at a possibly different target.

Greedy / deploy. The destination is $\arg \max_d \tilde{\ell}_{s,d}$ and the fire decision is $f_s^{\text{greedy}} = \mathbb{K}[p_s \geq 0.5]$ — a single concentrated strike per firing source. (Under the retired Dirichlet WHERE the deploy rule was a temperature-sharpened masked softmax, because the Dirichlet *mean* was concentration-blind and floored every $\lfloor \cdot \rfloor$ below MIN_LAUNCH_SHIPS , firing zero launches — a residual passivity bug; see Section 7.)

7 Why categorical and not Dirichlet (and how the earlier attempts blew up)

<i>Variable reference — this section</i>	
$A[s, \cdot] \in \Delta^{E-1}$	retired Dirichlet allocation row over the simplex (ship fractions)
$\alpha_{s,d}$	Dirichlet concentration, Eq. (12)
κ, ϵ	ALLOC_KAPPA concentration gain / Dirichlet floor; LOGIT_CLAMP= 8
$\ell_{s,d}$	bilinear logits — shared by both WHERE distributions
r	importance ratio $\exp(\log \pi_\theta - \log \pi_{\theta_{\text{old}}})$
KL, k_3	trust-region divergence and its low-variance estimator
$x_d, \Delta \log \pi_\theta(a)$	simplex coordinate; per-step log-prob movement of a stored action
$\nabla_\ell \log \text{softmax}(\ell)_d$	categorical policy gradient = $\text{onehot}(d) - \text{softmax}(\ell)$ ($\rightarrow 0$ when confident)

This section is the full account of the architecture decision that distinguishes v7+ from v6, including the trust-region collapse and the reward/loss blow-ups the user asked to have on record.

7.1 The Dirichlet WHERE (what v6 did)

v5/v6 made each owned source row a continuous **allocation simplex** $A[s, \cdot] \in \Delta^{E-1}$ — ship *fractions* over destinations including the self-diagonal (the HOLD share) — and placed a Dirichlet on it:

$$\alpha_{s,d} = \text{softplus}(\text{clip}(\ell_{s,d}, \pm \text{LOGIT_CLAMP})) \cdot \text{ALLOC_KAPPA} + \epsilon, \quad A[s, \cdot] \sim \text{Dirichlet}(\alpha_s), \quad (12)$$

with log-prob/entropy the per-source Dirichlet terms summed over owned planets. The intent was expressivity: one source could split its garrison across several destinations in a single turn.

7.2 The measured failure: warm-started PPO destroys the Dirichlet policy

Starting PPO from a BC clone (Section 10) with the Dirichlet head, under *all* guardrails of Section 9.2 (run `runs/v6/ppoS2`), the fixed-opponent gauntlet decayed $0.755 \rightarrow 0.625 \rightarrow 0.276 \rightarrow 0.104$ over 40 iterations while the per-update k_3 KL spiked to 2–9 *despite* a per-minibatch KL hard stop. This is not a tuning artifact; it is intrinsic to the distribution.

7.3 How the KL trust region silently collapses

PPO is an off-policy update made valid by an importance ratio $r = \exp(\log \pi_\theta(a) - \log \pi_{\theta_{\text{old}}}(a))$ and kept honest by a **trust region**: keep π_θ close in KL to the data-collecting $\pi_{\theta_{\text{old}}}$. Two mechanisms enforce it — the ratio *clip* ($\pm\epsilon$ around 1) and a KL *early-stop* — but **both act between optimiser steps**, the clip bounding each sample’s loss contribution and the stop bounding damage per epoch/minibatch. Neither bounds how far a *single gradient step* moves $\log \pi_\theta(a)$.

For a categorical, that per-step movement is small and bounded. For a Dirichlet sharpened by BC it is catastrophic. BC drives $\alpha = \text{softplus}(\ell) \kappa$ into near-one-hot rows, so the policy sits at a *vertex* of the simplex. The Dirichlet log-density there is steep — the $(\alpha_d - 1) \log x_d$ terms with suppressed $\alpha_d = \text{softplus}(-6) \approx 0.003 < 1$ are *singular* at exactly the corner every sample occupies, injecting $|\log x_d| \approx 14$ -scale contributions. A single clipped, LR-warmed step then moves a stored action’s log-prob by whole *nats*:

$$\Delta \log \pi_\theta(a) \sim \text{several nats} \Rightarrow r = e^\Delta \gg 1 \Rightarrow \text{KL} \approx \frac{1}{2}(\log r)^2 \text{ blows up to 2–9 in one update.}$$

Once r leaves $1 \pm \epsilon$ the surrogate gradient on that sample is gated off, but the damage is already done: the ratio is no longer a valid importance weight, the advantage is being applied through a distribution that no longer matches the data, and the BC behaviour erodes. The KL stop fires *after* the epoch that already ran away — exactly the “trust region collapses when the policy outcome moves too far” failure: the controllers all sit downstream of a per-step move they cannot see.

7.4 The replacement and why it is self-stabilising

The WHERE head keeps its bilinear logits $\ell_{s,d}$ but the distribution becomes a per-source **categorical** (Eq. (8)); the sampled row is the one-hot of the drawn destination, so the action bundle, decode, rollout buffers, league fingerprint, and checkpoints are all untouched. The categorical’s score $\log \text{softmax}(\ell)_d$ has *bounded* sensitivity in ℓ , and the policy gradient $\partial \log \text{softmax}(\ell)_d / \partial \ell = \text{onehot}(d) - \text{softmax}(\ell)$ **collapses to ≈ 0 on a confident sample** (when softmax is already near one-hot at the chosen d). A sharp categorical is therefore self-stabilising; a sharp Dirichlet cannot settle, because its gradient stays ≈ 5 *forever* (the singular corner terms above).

Quantified (CPU probe, scripts/probe_dist_sensitivity.py). Over $E=48$ logit rows, identical logits, sampled actions: the Dirichlet’s $\|\partial \log \pi(a) / \partial \ell\|_2$ is $5\text{--}6\times$ the categorical’s at every sharpness regime, and the log-prob movement per unit-direction logit perturbation is $5\text{--}10\times$ — which compounds *quadratically* in KL, matching the $\sim 100\times$ KL gap measured between the real runs. The decisive asymmetry is at BC sharpness: the categorical’s gradient on confident samples $\rightarrow 0$ while the Dirichlet’s stays ≈ 5 . Measured end-to-end: the first-update KL drops from $0.53\text{--}2.06$ (Dirichlet) to $0.008\text{--}0.010$ (categorical), same setup.

7.5 Two more independent reasons

1. **Deploy equivalence.** The *deployed* Dirichlet already collapsed to a temperature-sharpened softmax over the same ℓ — effectively one-hot. The continuous “split allocation” expressivity was paid for in training fragility and *never used* at deploy; the gate-as-rate already delivers multi-target pressure over time (Section 6).
2. **Search-readiness.** A fully discrete factored action (categorical WHERE $\times$ Bernoulli FIRE) is the prerequisite for the planned Gumbel-AlphaZero extension (Appendix A); the Dirichlet was the blocker.

7.6 Attribution: what the Dirichlet did and did not cause

A 2×2 (from-scratch $\times$ {categorical, Dirichlet}, full guardrails) plus a low-LR BC leg separates three phenomena previously conflated:

phenomenon	cause	evidence
bootstrap wall (can’t beat starter from scratch in reasonable time)	<i>not</i> the distribution: credit assignment + opponent gap	scratch-cat gauntlet 0.083, Elo 170; scratch-dir 0.016, Elo 125 (both walled)
first-update slam (launch rate annihilated at it1)	Dirichlet $\times$ <i>unguarded</i> update; guardrails alone remove it	old code: it1 KL 2.06, lnc 13.5 $\rightarrow$ 1.0; guardrailed dir: KL ≤ 0.024 , no slam
post-BC erosion	Dirichlet density sharpness; LR-dependent (trust-region misfit)	LR 10^{-4} : gauntlet 0.755 $\rightarrow$ 0.104; LR 10^{-5} : 0.719 $\rightarrow$ 0.771 (survives)

The bootstrap wall is *distribution-independent* (BC removes it under either; the v5 actor’s destination head was already categorical and still needed 1400 iterations for a 0.47 starter win rate). The

Dirichlet’s real crime is the update fragility, and even that is survivable at $10\times$ reduced LR — but strictly dominated: BC+Dirichlet at LR 10^{-5} reaches gauntlet 0.771 with median KL 0.102 (still $2\text{--}3\times$ over target), vs. BC+categorical at LR 10^{-4} reaching 0.812 with KL 0.035 and Elo +161 over the same 40 iterations. You can pay for the Dirichlet with a $10\times$ smaller trust region; there is no reason to.

7.7 The earlier reward/loss blow-ups (full list)

The categorical fix removed the *sharpness* blow-up; for the record, these are the other divergences that earlier designs hit and the guards that contained them:

1. **Continuous squashed-Gaussian actor.** Summing the squashed-Gaussian log-prob over $E\cdot 3K$ components, an extreme mean μ with a saturated sample (latent far from μ) over a tiny σ^2 produces a huge $\log \mathcal{N}$, hence an exploding step and $\exp(\log p - \log p_{\text{old}}) \rightarrow \infty$. Guards: tight log-std range $[-2, 1]$, a μ clamp inside the density, a log-ratio clamp before exp, a $[-10, 10]$ clamp on the whitened advantage (a near-degenerate group otherwise explodes it), and skipping any optimiser step with non-finite gradient.
2. **Frozen policy / never exploits.** The entropy bonus pushed σ to its cap, the sampled action stayed noise-dominated, the per-trajectory advantage decorrelated from the mean, and the policy-gradient terms cancelled ($\text{adv_std} = 1$ yet $\text{KL} \rightarrow 0$). Fixed by a two-phase σ -cap schedule (force the cap down, then release to a state-dependent head).
3. **Catastrophic first update (from scratch).** Iteration-1 KL 2.06, clip-fraction 0.98 — every sample clipped, launch rate $13.5 \rightarrow 1.0$ per step in one iteration. The v6 end-of-epoch KL check fires only *after* a full epoch ran away. Fixed by the per-minibatch hard stop + LR warmup + critic-only warmup (Section 9.2).
4. **Dirichlet entropy bonus.** Differential entropy of a peaked Dirichlet is unboundedly *negative* (≈ -4000 per source after BC), so with coefficient 0.005 the “bonus” became a +20 *loss* term whose gradient actively blurred the BC-learned WHERE head (measured at warmup-end: grad-norm 1380, KL 4.8, Elo $1032 \rightarrow 955$). Fixed: gate-only entropy (Section 9.2), keeping only bounded terms.
5. **Value-target dynamic range.** The MSE critic target spans the $\pm 1000/\sigma_R$ terminal spike against $O(10^{-2})$ per-step shaping, so an un-normalised value loss dominates and destabilises the shared trunk. Fixed by PopArt (Section 9.1).

8 Reward: N -player potential shaping and outcome

Variable reference — this section	
$S^{(p)}$	player p total ships (planets + fleets)
$\Pi^{(p)}, \Pi_{\text{tot}}$	owned production / total alive production
$\ell(n)$	ship-log $\ln(1+n)/\ln 1000$
$\Phi_{\text{ship}}, \Phi_{\text{prod}}$	ship- and production-potentials vs. the strongest opponent, $\in [-1, 1]$
r_t	per-step potential-shaping reward, Eq. (14)
$c_{\text{ship}}, c_{\text{prod}}$	shaping coefficients, <code>SHAPE_SHIP=SHAPE_PROD= 50</code>
γ	discount (the <i>same</i> γ as GAE)
$F(s, s')$	shaping term $\gamma\Phi(s') - \Phi(s)$; Φ a potential (telescopes $\Rightarrow$ policy-invariant)
sc_k	pairwise terminal score vs. seat k , $\in \{0, \frac{1}{2}, 1\}$
Ω	placement-linear outcome $\in \{-1, -\frac{1}{3}, +\frac{1}{3}, +1\}$, Eq. (15)
O, W, L	terminal payment $\Omega \cdot (\cdot)$; win/loss magnitude $W=L=1000$
σ_R	<code>PPO_REWARD_SCALE= 100</code> (reward scale before GAE)

The reward is potential-based shaping + a terminal outcome, generalised so that $N=2$ reproduces the two-player form exactly. The earlier four-channel *event* reward (outcome + capture + dispatch + production-milestone) was **retired** because its dense channels were farmable — capture rewarded border-flip churn, dispatch rewarded ship expenditure, milestones rewarded hoarding, all maximisable while *losing*.

Potentials vs. the strongest opponent. With per-player totals $S^{(p)}$ (planets + fleets), owned production $\Pi^{(p)}$, total alive production Π_{tot} , and ship-log $\ell(n) = \ln(1+n)/\ln 1000$,

$$\Phi_{\text{ship}}(s) = \ell(S^{(0)}) - \ell\left(\max_{p \geq 1} S^{(p)}\right), \quad \Phi_{\text{prod}}(s) = \frac{\Pi^{(0)} - \max_{p \geq 1} \Pi^{(p)}}{\max(\Pi_{\text{tot}}, 1)}. \quad (13)$$

The max is the right opponent aggregate for a win-the-game potential: in an FFA you are measured against whoever currently leads, not the average. Both are bounded in $[-1, 1]$ and reduce to the two-player pair at $N=2$.

Shaping reward. The per-step reward is the potential difference

$$r_t = c_{\text{ship}}(\gamma\Phi_{\text{ship}}(s_{t+1}) - \Phi_{\text{ship}}(s_t)) + c_{\text{prod}}(\gamma\Phi_{\text{prod}}(s_{t+1}) - \Phi_{\text{prod}}(s_t)), \quad (14)$$

with $c_{\text{ship}} = c_{\text{prod}} = \text{SHAPE_SHIP} = \text{SHAPE_PROD} = 50$ and the *same* γ as GAE.

Policy invariance (why this is un-hackable). For $F(s, s') = \gamma\Phi(s') - \Phi(s)$ the added discounted return telescopes, $\sum_{t \geq 0} \gamma^t F(s_t, s_{t+1}) = \gamma^T \Phi(s_T) - \Phi(s_0)$, which depends only on the fixed start and terminal, not the policy (Ng, Harada & Russell, 1999). So the optimal policy and the ordering of policies are unchanged versus training on the sparse outcome alone, while the gradient is now dense and win-aligned at every step. A quantity that returns to its prior value contributes net zero — no reward-farming basin exists.

Pairwise scores and the 4p outcome. At termination, against each seat k with total $S^{(k)}$:

$$sc_k = \begin{cases} 1 & S^{(0)} > S^{(k)} \\ \frac{1}{2} & S^{(0)} = S^{(k)} \\ 0 & S^{(0)} < S^{(k)}, \end{cases} \quad \Omega = \begin{cases} \text{sign}(S^{(0)} - \max_{p \geq 1} S^{(p)}) & N=2 \text{ or } \text{OUTCOME_4P}=\text{winner}, \\ 2 \cdot \frac{1}{N-1} \sum_k sc_k - 1 & \text{OUTCOME_4P}=\text{placement (default)}. \end{cases} \quad (15)$$

The placement-linear $\Omega \in \{-1, -\frac{1}{3}, +\frac{1}{3}, +1\}$ maps finishing place affinely (beat-all = +1, lose-to-all = -1): a dense, correctly-ordered signal where the official top-score-only rule pays a 4p game like a lottery (2nd of 4 = dead loss). The terminal payment is the win/loss magnitude scaled linearly by Ω , $O = \Omega \cdot (\Omega > 0 ? W : L)$ with $W=L=1000$, so 2p (where $\Omega \in \{\pm 1, 0\}$) pays as a ± 1000 win/loss. The per-seat sc_k are also the dual-Elo update signal (Section 11); the logged 4p `win_rate` is the strict “beat *every* opponent” indicator $\mathbb{K}[\min_k sc_k \geq 1]$.

Scaling into GAE. Both r_t and the scattered O are divided by $\sigma_R = \text{PPO_REWARD_SCALE} = 100$ before GAE (the outcome is placed on the episode’s last active step so GAE propagates it backward); advantages are re-normalised afterward. Empirically the outcome dominates the per-game return (e.g. $O \sim 450$ vs. shaping ~ 8 each), the intended regime.

9 PPO + GAE update

Variable reference — this section

$V(s), \delta_t$	value $\sigma\tilde{V}_\theta + \mu$; TD residual $r_t + \gamma V(s_{t+1}) - V(s_t)$
$\hat{A}_t^{\text{GAE}}, \hat{R}_t$	GAE advantage and its return target $\hat{A}_t^{\text{GAE}} + V(s_t)$
$\mathcal{L}^V$	(normalised) value loss
γ, λ	discount 0.997, GAE decay 0.98
r_t, ϵ	importance ratio; clip radius <code>CLIP</code> = 0.2
$\mathcal{L}^{\text{CLIP}}, \mathcal{L}$	clipped surrogate; total loss $\mathcal{L}^{\text{CLIP}} + c_V \mathcal{L}^V - c_H \bar{\mathcal{H}}$
$c_V, c_H, \bar{\mathcal{H}}$	<code>vf_coef</code> = 0.5; entropy coefficient; per-source mean entropy
k_1, k_2, k_3	Schulman KL estimators of $\text{KL}(\pi_{\theta_{\text{old}}} \parallel \pi_\theta)$
<code>approx_kl, clipfrac</code>	logged k_1 -form KL and clip fraction $\mathbb{E}[\mathbb{I}(r_t - 1 > \epsilon)]$
μ, ν, σ, β	PopArt running mean, second moment, std, EMA rate $\beta = 3 \times 10^{-4}$
$\hat{A}_i, \mu_j, \varsigma_j$	(GRPO, retired) group-centred advantage, group mean/std; π_{ref} frozen reference

The algorithm is actor-critic PPO with Generalized Advantage Estimation. With the value head $V(s) = \sigma\tilde{V}_\theta(s) + \mu$, the TD residual $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, and the alive/done masks,

$$\hat{A}_t^{\text{GAE}} = \sum_{l \geq 0} (\gamma\lambda)^l \delta_{t+l}, \quad \hat{R}_t = \hat{A}_t^{\text{GAE}} + V(s_t), \quad \mathcal{L}^V = \frac{1}{2} (\tilde{V}_\theta(s_t) - (\hat{R}_t - \mu)/\sigma)^2, \quad (16)$$

with $\gamma = \text{GAMMA} = 0.997$ and $\lambda = \text{GAE_LAMBDA} = 0.98$ (a ~ 500 -step game with $\gamma = 0.99$ discounts the terminal to $\approx 7 \times 10^{-3}$ and confines GAE credit to the last $\sim 1/(1-\gamma\lambda)$ steps; the higher values let the win-aligned signal reach mid-game actions). The `done` flag is set at *any* terminal (death *or* time cap), so a time-capped env does not add a spurious $\gamma V(s_T)$ bootstrap on top of the scattered outcome; a true death contributes no bootstrap.

Clipped surrogate and total loss. With the per-sample ratio $r_t = \exp(\log \pi_\theta - \log \pi_{\theta_{\text{old}}})$,

$$\mathcal{L}^{\text{CLIP}} = -\mathbb{E}_t \left[\min(r_t \hat{A}_t^{\text{GAE}}, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \hat{A}_t^{\text{GAE}}) \right], \quad \epsilon = \text{CLIP} = 0.2, \quad (17)$$

$$\mathcal{L} = \mathcal{L}^{\text{CLIP}} + c_V \mathcal{L}^V - c_H \bar{\mathcal{H}}, \quad c_V = \text{vf_coef} = 0.5, \quad (18)$$

where $\log \pi_\theta$ is the per-source sum of Eq. (10). Optimisation: collect on-GPU under $\pi_{\theta_{\text{old}}} = \pi_\theta$, compute GAE, then `UPDATE_EPOCHS` = 3 passes over shuffled minibatches with Adam (global grad-norm clip `MAX_GRAD_NORM`), recomputing r_t each minibatch.

KL: the trust region and its estimators. For discrete distributions $\text{KL}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$ (equality iff $p=q$; not symmetric), and the factored policy’s joint KL is the *sum* of the per-source factor KLs over owned planets. PPO controls the old-vs-new KL implicitly through clipping; the loop monitors it with Schulman’s estimators of $\text{KL}(\pi_{\theta_{\text{old}}} \parallel \pi_\theta)$ (writing $\log r = \log \pi_\theta - \log \pi_{\theta_{\text{old}}}$):

$$k_1 = \mathbb{E}_{\pi_{\theta_{\text{old}}}} [-\log r] \quad (\text{unbiased; high variance; can be negative}), \quad (19)$$

$$k_2 = \frac{1}{2} \mathbb{E}_{\pi_{\theta_{\text{old}}}} [(\log r)^2] \quad (\text{low variance; biased; } \geq 0), \quad (20)$$

$$k_3 = \mathbb{E}_{\pi_{\theta_{\text{old}}}} [(r-1) - \log r] \quad (\text{unbiased; low variance; } \geq 0). \quad (21)$$

k_3 is the control-variate fix: $(r-1)$ has zero expectation under $\pi_{\theta_{\text{old}}}$ and $r-1 \geq \log r$ makes every term ≥ 0 . The loop logs `approx_kl` = $\text{mean}(\log \pi_{\theta_{\text{old}}} - \log \pi_\theta)$ (the k_1 form) and the clip fraction `clipfrac` = $\mathbb{E}[\mathbb{I}(|r_t - 1| > \epsilon)]$. `approx_kl` and `clipfrac` rise together: KL growth is what activates the clip, and a spiking clipfrac with a large approx-KL is the signal the trust region is binding.

9.1 PopArt value normalisation

The MSE critic target spans a huge dynamic range (Section 7), so the value head is trained in a *normalised* space with output-preserving rescaling (Hessel et al., 2018). With running statistics over GAE returns $\hat{R}$, EMA rate $\beta = \text{POPART_BETA} = 3 \times 10^{-4}$,

$$\mu \leftarrow (1-\beta)\mu + \beta \mathbb{E}[\hat{R}], \quad \nu \leftarrow (1-\beta)\nu + \beta \mathbb{E}[\hat{R}^2], \quad \sigma = \sqrt{\nu - \mu^2}. \quad (22)$$

The value loss regresses the normalised target $\mathcal{L}^V$ above, while GAE/bootstrapping use the de-normalised $V = \sigma \tilde{V}_\theta + \mu$. On each statistics update $(\mu, \sigma) \rightarrow (\mu', \sigma')$ the final value layer is rescaled to *preserve* raw outputs, $W \leftarrow W \sigma / \sigma'$, $b \leftarrow (\sigma b + \mu - \mu') / \sigma'$, so $V(s)$ is invariant to the update (verified to 4×10^{-7}). PopArt state is persisted in the resumable train-state.

9.2 The three PPO guardrails

These make the post-BC updates structurally incapable of destroying the warm start:

1. **Per-minibatch KL hard stop** (KL_STOP_MINIBATCH). The update breaks *mid-epoch* the moment a minibatch’s k_3 KL exceeds $\text{KL_HARD_MULT} (1.5) \times \text{KL_TARGET} (0.05)$. The end-of-epoch check provably cannot prevent the iteration-1 slam (KL 2.06 happened *within* one epoch).
2. **Linear LR warmup** (LR_WARMUP_ITERS): $\eta_t = \eta \cdot \min(1, (t+1)/W)$ over the first W iterations of a *train()* call, so every fresh-Adam phase (from-scratch, post-BC, resumed-with-fresh-moments) ramps in.
3. **Critic-only warmup** (VALUE_WARMUP_ITERS): for the first K iterations the policy surrogate and entropy are multiplied by 0 *and only val_* parameters are stepped*. The second clause is load-bearing — with a shared trunk, the value loss alone (backpropagating into the trunk) dragged the BC policy by KL 0.15–0.84 per iteration with the surrogate fully off. Rationale: after BC the actor is strong but the critic is random, so advantages against a random baseline are structured noise. With the val-only restriction the warmup measures $\text{KL} \equiv 0.000$, then the actor unfreezes against a usable baseline.

9.3 Entropy bonus: gate-only

The bonus keeps only *bounded* terms: the gate Bernoulli entropy ($\leq \ln 2$) plus the categorical WHERE entropy ($\leq \ln E$) weighted by ENT_DIR_COEF (default 0). The coefficient on the gate term is annealed $\text{ENT_COEF_START} = 0.02 \rightarrow \text{ENT_COEF_END} = 0.002$ (explore $\rightarrow$ exploit), and is normalised *per owned source*, $\bar{\mathcal{H}} = \frac{1}{|\mathcal{B}|} \sum_b \mathcal{H}_b / \max(n_{\text{owned}}^b, 1)$. Exploration is supplied by the stochastic gate, categorical sampling, and PFSP opponent diversity — not by fighting the BC sharpness (the unbounded-negative Dirichlet entropy was a blow-up source, Section 7).

9.4 GRPO (the retired alternative)

The original algorithm had no critic: it drew N groups, each fixing a start world and opponent and rolling out G trajectories under $\pi_{\theta_{\text{old}}}$, and centred each return against its group, $\hat{A}_i = (R_i - \mu_j) / (\varsigma_j + \varepsilon)$ (or $R_i - \mu_j$ unwhitened), broadcast to all of the trajectory’s timesteps. It used the same clipped surrogate and a KL anchor $\beta \mathbb{E}[k_3(\pi_\theta \| \pi_{\text{ref}})]$ to a frozen reference. PPO + GAE superseded it because the group baseline collapses each trajectory to a single episode return (no temporal credit), which is what produced the passivity wall (Section 1).

10 BC warm-start in the gated-categorical action space

<i>Variable reference — this section</i>	
$\text{fire}_s^*, \text{where}_s^*$	medium-expert fire label / destination one-hot; d_s^* the target index
S_s	source garrison (the expert fires when $\lfloor S_s/2 \rfloor \geq 20$)
$p_s, \tilde{\ell}_{s,\cdot}$	smooth gate probability / masked WHERE logits
O, F	set of owned sources / set of firing sources (this F is a <i>set</i> , not the feature count)
$\mathcal{L}_{\text{BC}}$	BC loss = gate BCE (all owned) + WHERE cross-entropy (firing only)

The bootstrap — not the algorithm — was the wall (Section 1). BC removes it by starting PPO from a policy that already commits, aims, and wins against the scripted ladder. This is the payoff of the gated-allocation action space: the strongest scripted bot maps *exactly* onto the two heads.

The expert, native to the heads. The clone target is **medium** (nearest non-owned planet, lead-aimed, fire when $\lfloor S_s/2 \rfloor \geq 20$):

$$\text{fire}_s^* = \mathbb{1}[\text{owned}_s \wedge \text{alive}_s \wedge \lfloor S_s/2 \rfloor \geq 20 \wedge \exists \text{target}], \quad \text{where}_s^* = \text{onehot} \left(\arg \min_{d: \text{non-owned}} \|x_d - x_s\| \right).$$

Two deliberate deviations: the gated decode dispatches the *full* garrison on fire (the action space cannot “send half”), so the clone is “full-send medium” — measured *stronger* than its teacher; and sun-occluded targets are dropped from the WHERE loss (“medium minus sun-suicide”).

Loss (native to the distribution). With p_s the smooth gate probability and $\tilde{\ell}_{s,\cdot}$ the masked WHERE logits,

$$\mathcal{L}_{\text{BC}} = \underbrace{-\frac{1}{|O|} \sum_{s \in O} [\text{fire}_s^* \log p_s + (1 - \text{fire}_s^*) \log(1 - p_s)]}_{\text{gate BCE, all owned sources}} - \underbrace{\frac{1}{|F|} \sum_{s \in F} \log \text{softmax}(\tilde{\ell}_{s,\cdot})_{d_s^*}}_{\text{WHERE CE, firing sources only}}.$$

The CE *is* the categorical log-likelihood (Eq. (8)), so supervised and RL phases optimise the same parameters with consistent semantics — no new heads, no adapter. (It is also the exact log-likelihood the Dirichlet’s softmax-deploy approximated, so a single BC checkpoint warm-starts either WHERE variant.)

Data: states the clone itself visits. Rollouts are generated by the *expert acting in the clone’s own action space* against a starter-heavy scripted mix, so the state distribution BC fits is the one the warm-started policy will visit — a free DAGger-like property (no compounding-error gap on the states that matter).

Result and quality gate. A 256/L2 clone in ~ 144 s reaches greedy-deploy GpuEnv scores 1.000/0.938/0.625/0.875 vs. **random/starter/greedy/medium** — league Elo ≈ 900 –940, versus the best-ever from-scratch RL of 0.47 vs. starter at ~ 20 h. A deliberately undertrained clone (64-dim, 2 rounds) reached only dest-acc 0.40 and scored 0.03 vs. starter: **BC inits are trustworthy only above dest-acc ≈ 0.85** . The decisive A/B (same BC init, same guardrails, same seeds, only the WHERE distribution changed): categorical climbs $0.76 \rightarrow 0.81$ gauntlet, Elo $938 \rightarrow 1099$, KL pinned in $[0.025, 0.045]$ for all 40 iters; Dirichlet erodes $0.755 \rightarrow 0.104$ with KL spiking 0.2–8.9. The recipe transfers untouched across a $7\times$ parameter jump (256/L2 $\rightarrow$ 512/L4), and the rollout is simulator-bound so model capacity is nearly free.

11 The mixed 2p/4p league

<i>Variable reference — this section</i>	
e^2, e^4	dual correlated Elo (2p / 4p) carried by every member
$\mathbb{E}, \mathbb{E}_k^{(4)}$	logistic win expectation (2p / vs. seat k in 4p)
K, K_4, κ	$\text{ELO_K}=\text{ELO_K4}=32$; cross-format coupling $\kappa=0.25$
$\Delta, \bar{s}, \text{sc}_k$	rating update; mean learner score; pairwise score vs. seat k
s_4, a	scheduled 4p match share ($\text{S4_BASE}/\text{MIN}/\text{MAX}/\text{GAIN}$); error-diffusion accumulator
w_m, p_m, f	PFSP draw weight; logistic expectation vs. member m ; PFSP curve
JS	Jensen–Shannon behavioural-footprint (anti-clone) distance
β, w	weak-point boost factor; learner’s actual score EMA
$\hat{r}, R, \bar{a}$	recalibration back-solved rating; anchor rating; mean anchored 4p rating
g, g_2, g_4, w_4	mixed best-ckpt gauntlet = $(1-w_4)g_2 + w_4g_4$, $w_4=0.5$

A single training program schedules both formats, because heads-up strength and FFA strength are correlated but not identical (the `it1041` post-mortem: 1.000 heads-up gauntlet, yet 29% 4p top-rate vs. 58%).

Unified rollout. One `collect_ppo(env, net, pool, ..., seats, n_players)` runs both formats. The learner is always player 0 with on-GPU buffers and GPU GAE; each opponent seat is either a scripted anchor (on-GPU, zero forwards) or a frozen neural member acting per-step under `no_grad`. A 4p iteration costs the learner’s rollout + up to three no-grad forwards per step; $\text{B_4P} = B$ keeps the learner’s transition count format-independent.

Dual correlated Elo. Every member — and the learner — carries two ratings (e^2, e^4); scripted anchors and invited members are anchored in both (never move); snapshots float. A 2p iteration vs. member m with mean learner score $\bar{s}$ and expectation $\mathbb{E} = (1 + 10^{(e_m^2 - e_\ell^2)/400})^{-1}$ updates

$$\Delta = K(\bar{s} - \mathbb{E}), \quad e_\ell^2 += \Delta, \quad e_\ell^4 += \kappa\Delta, \quad (23)$$

($K = \text{ELO_K} = 32$, coupling $\kappa = \text{ELO_COUPLING} = 0.25$, non-anchor m mirrored). A 4p FFA becomes $N-1$ pairwise updates at $K_4/(N-1)$ each (the standard multiplayer decomposition; total motion per game matches a 2p game at K_4):

$$\Delta_k = \frac{K_4}{N-1}(\text{sc}_k - \mathbb{E}_k^{(4)}), \quad e_\ell^4 += \Delta_k, \quad e_\ell^2 += \kappa\Delta_k \text{ for each } k, \quad (24)$$

$K_4 = \text{ELO_K4} = 32$. The coupling encodes that the two skills share most of their variance (a 2p result moves the 4p belief a quarter as much, and vice versa) without forcing them equal — it is exactly their *gap* that drives the scheduler.

Gap-driven match mix. The 4p share is a bounded affine function of the dual-rating gap,

$$s_4 = \text{clip}\left(\text{S4_BASE} + \text{S4_GAIN} \cdot \frac{e_\ell^2 - e_\ell^4}{400}, \text{S4_MIN}, \text{S4_MAX}\right), \quad (25)$$

so when the 4p rating lags, more 4p is scheduled (within a 1:1 .. 1:6 envelope). The scheduler is **error-diffused**: an accumulator $a += s_4$ each iteration plays 4p iff $a \geq 1$ (then $a -= 1$), so the realized ratio equals the target exactly over any window (a Bernoulli draw would only match in expectation, with format droughts at small counts).

Membership and matchmaking. Opponents are PFSP draws over all pool members m with weight $w_m = \text{MATCH_ELO_W} f(p_m) + \text{MATCH_FP_W} \text{JS}(\hat{\pi}_m, \hat{\pi}_\ell) + \text{PFSP_FLOOR}$, where p_m is the logistic expectation, f the PFSP curve ($p(1-p)$ “even” or $(1-p)^P$ “hard”), and the JS term is an *anti-clone* behavioral-footprint distance (down-weights near-identical snapshots, keeps the maximally-distinct scripted anchors in the mix). The footprint is computed on cached *real* encoded states. 4p seats are $k=3$ weighted draws without replacement. Two additions:

- **Invited members.** The top $\text{INVITE_MAX} = 4 \cdot \text{pt}$ by anchored rating from INVITE_DIRS join as fixed high-level bots; the anchor resolves as filename suffix `_elo<NNNN> > ckpt-meta elo > INVITE\_DEFAULT\_ELO = 750` (the submitted `it1041` level). While unmastered they sample at $\times \text{INVITE_MATCH_BOOST} = 2$; once the learner’s per-format EMA passes $\text{INVITE_MASTER_WR} = 0.75$ their rate drops to $\times \text{INVITE_MASTERED_W} = 0.2$ (a small anti-forgetting floor). Legacy ($F=101$) members get the obs adapter of Section 4.
- **Weak-point matchmaking.** Per member and format, compare the learner’s actual score EMA w against the Elo-expected $\mathbb{E}$; when reality runs below theory, boost the weight $\beta = \min(\text{WEAK_BOOST_MAX}, 1 + \text{WEAK_BOOST_GAIN}(\mathbb{E} - w))$, gated by a minimum game count and a deadband (so EMA noise does not trigger it). It fires only on underperformance.

Curriculum gate. Strong anchors are withheld from a weak learner. When the learner’s online Elo first reaches $\text{ADVANCE_TRIGGER_ELO} = 525$, a full recalibration with all anchors on fresh seeds runs; if it grades $\geq \text{ADVANCE_CONFIRM_ELO} = 425$, the advanced anchors unlock permanently (the gate refuses to be fooled by self-play inflation).

Recalibration (removes self-play inflation). Online Elo inflates as the learner farms under-rated snapshots. Every ELO_RECAL_EVERY iters on fresh worlds, the learner’s rating is re-grounded against the fixed anchors via the logistic back-solve: for measured score s (clamped to $[0.02, 0.98]$) vs. an anchor of rating R , $\hat{r} = R + 400 \log_{10} \frac{s}{1-s}$, averaged over the active anchors. The 4p analog re-grounds e^4 from the fixed FFA trio `{starter, greedy, intermediate}` with mean anchored rating $\bar{a}$: $e^4 = \bar{a} + 400 \log_{10}(s/(1-s))$.

Mixed best-checkpoint gauntlet. “Best” is selected on the mix it will be graded on, not 2p alone (the 2p-only rule is what crowned `it1041` while it lost FFAs):

$$g = (1 - w_4) g_2 + w_4 g_4, \quad w_4 = \text{GAUNTLET_4P_W} = 0.5, \quad (26)$$

g_2 the 2p gauntlet (greedy play vs. `starter/intermediate/greedy` on held-out worlds), g_4 the mean pairwise score (Eq. (15) left) of a 4p FFA vs. the same trio on the same worlds.

Scripted anchors and the leaderboard-calibrated scale. The pool is pinned by hand-coded bots, all vectorised in the GPU env. The anchor Elos were re-pinned from the in-house round-robin to the *submitted* Kaggle levels so the learner’s rating reads against the competition:

anchor	Elo	behaviour
random	0	random heading, half-garrison (≥ 20) from any planet
starter	450	half-garrison (≥ 20) at the nearest <i>static</i> non-owned planet, straight aim
intermediate	475	in-flight-aware capture from the nearest capable source, lead-aimed; local rebalancing; comet-escape
greedy	500	every owned planet captures the nearest beatable non-owned planet within radius, lead-aimed (strong)
medium	500	“starter++”: nearest non-owned <i>including</i> rotating, lead-aimed, half-garrison

Empirical strength order is **greedy** > **medium** > **intermediate** > **starter** > **random**. The compressed scale understates in-house win probabilities (starter→medium is a $\sim 90\%$ -win gap but is now 50 Elo $\approx 57\%$), so PFSP/weak-boost expected scores flatten toward $\frac{1}{2}$; the weak-point detector compensates since it compares against the same $\mathbb{E}$. League state stores member ratings, so resuming a pre-rescale train-state keeps the old anchors — fresh runs use the new ones.

12 Deployment

<i>Variable reference — this section</i>	
F	body-feature count; v9 checkpoints are $F=104$ (do not load in v7-era tools)
<code>n_players</code>	set to 4 in FFA episodes, 2 heads-up; the adapter maps each opponent to its own owner 1..3
B, E	inference batch ($B=1$) and planet-token count per move
d, L	served model width and depth; cost $\approx 12d^2L \cdot E$ MACs/move

Seat-preserving adapter (the v9 constraint). The v7-era Kaggle adapters map *every* opponent to the single enemy owner before encoding — correct for v7/v8 (that *is* their training view) and **wrong for v9**: it would zero the seat channels and discard the trained per-enemy behaviour. The v9 submission adapter must *preserve seats*: map each opponent id to its own owner 1..3 (any fixed assignment — Eq. (4) is ego-relative, so only consistency within a game matters), set `n_players`= 4 in 4p episodes and = 2 heads-up. v9 checkpoints ($F=104$) do not load in the v7-era eval tools. The adapter is parity bit-exact across all egos.

Inference budget. The served agent runs one forward per turn on a single state ($B=1$, E planet tokens) on a 2-vCPU Kaggle box (~ 0.032 TFLOP/s fp32). The transformer cost is $\approx 12d^2L \cdot E$ MACs; even a 768/6 (43.8M-param) model is ~ 127 ms/move and a 1024/8 (103M) ~ 302 ms/move — both under the 1s `actTimeout`. **The deploy budget never binds the architecture choice up to ~ 100 M params**; model scale is bounded by training compute, not inference. The fp32 submission size ceiling (≤ 100 MB) is the real cap (≤ 26 M params fp32; a 512-wide $B=64$ trunk at 37.5M needs fp16).

13 Verification battery

- `scripts/test_v8_parity.py`: v7 vs. v8, 70 ticks (comet spawn included), identical random gated-alloc actions vs. scripted **medium** — planet/fleet state **and** encodings bit-exact (v8 is a strict superset of v7 on the 2p path).
- `scripts/test_v9_parity.py`: v8 vs. v9 — 2p *state* bit-exact; encodings bit-equal on all shared features; the one-hot exactly consistent with the scalar (self $\leftrightarrow$ 1, enemy $\leftrightarrow$ 2); channels 2/3 provably silent in 2p.
- `scripts/verify_enemy_id.py`: on a 4-seat state, v8 reads all enemies as code 2 (by design); v9 reads 0100/0010/0001 and is rotation-equivariant from every ego seat.
- `scripts/probe_dist_sensitivity.py`: the categorical-vs-Dirichlet gradient/log-prob sensitivity probe of Section 7.
- `scripts/smoke_v8_nb.py` / `scripts/smoke_v9_nb.py`: the full notebooks at SMOKE scale (BC $\rightarrow$ mixed 2p/4p league $\rightarrow$ resume): both formats scheduled within the envelope, KL inside the guardrail band, dual ratings and the mixed gauntlet logged; v9 additionally asserts $F=104$, the legacy wrapper on $F=101$ invites, and wrapper survival across the resume round-trip.

14 Config $\leftrightarrow$ symbols

Each section above opens with a local *variable reference* sheet defining the symbols it uses; this table is the master config-name $\leftrightarrow$ symbol/role map with deployed values.

config	symbol / role	value
<i>observation / network</i>		
N_BODY_FEATURES, F	ownership one-hot (v9)	14, $F=104$ (v8: 11, 101)
HIDDEN, d	trunk width	512
N_RES_BLOCKS / N_TX_LAYERS	trunk depth	6 / 4–5
USE_ATTENTION, USE_GLU, D_G	trunk A/B, globals dim	on, on, 32
DEST_HEAD	WHERE head, Eq. (7)	bilinear
INIT_DEST_SCALE, INIT_GATE_BIAS	calm init	0.02, -1
WHERE_DIST	WHERE distribution	categorical (dirichlet A/B)
GATE_TRIM_LO/HI, GATE_EPS	deploy trim / train clamp	0.2, 0.8 / 0.02
ALLOC_KAPPA, LOGIT_CLAMP	Dirichlet-only (A/B)	— / 8
<i>reward</i>		
SHAPE_SHIP, SHAPE_PROD	$c_{\text{ship}}, c_{\text{prod}}$, Eq. (14)	50, 50
OUTCOME_4P	Ω rule, Eq. (15)	placement (winner = official)
PPO_REWARD_SCALE, σ_R	GAE reward scale	100
<i>PPO + GAE</i>		
GAMMA, GAE_LAMBDA	γ, λ	0.997, 0.98
CLIP, vf_coef	ϵ, c_V	0.2, 0.5
UPDATE_EPOCHS	PPO epochs/iter	3
KL_TARGET, KL_HARD_MULT	per-minibatch hard stop	0.05, 1.5
LR_WARMUP_ITERS, VALUE_WARMUP_ITERS	guardrails	8, 5
ENT_COEF_START/END, ENT_DIR_COEF	gate / WHERE entropy	0.02 $\rightarrow$ 0.002, 0
USE_POPART, POPART_BETA	value normalisation β	on, 3×10^{-4}
<i>league</i>		
FOURP_ENABLED, B_4P	4p training, envs per 4p iter	on, B
S4_BASE/MIN/MAX/GAIN	match mix	0.60 / 0.50 / 6/7 / 1.0
ELO_K, ELO_K4, ELO_COUPLING	K, K_4, κ	32, 32, 0.25
INVITE_MAX/DEFAULT_ELO	invited members	4 / 750
INVITE_MATCH_BOOST/MASTER_WR/MASTERED_W	sparring focus	2.0 / 0.75 / 0.2
WEAK_BOOST_MIN_GAMES/MIN_GAP/GAIN/MAX	weak-point boost	5 / 0.05 / 10 / 3
MATCH_ELO_W/FP_W, PFSP_FLOOR	matchmaking blend	0.7 / 0.3 / 0.05
GAUNTLET_4P_W, w_4	mixed best-ckpt weight	0.5
ELO_STARTER/INTERMEDIATE/GREEDY/MEDIUM	anchor ratings	450 / 475 / 500 / 500
ADVANCE_TRIGGER_ELO/CONFIRM_ELO	curriculum gates	525 / 425
MIN_LAUNCH_SHIPS, $r_{\odot}$	decode floor; sun radius	1; 10

A Appendix: search-based improvement (the planned successor)

Variable reference — this section

m	number of joint actions sampled and searched (degrades to PPO at $m=1$)
g	per-factor Gumbel noise (Gumbel-Top- k sampling)
$\hat{q}$	completed action-value (a depth-1 real env step bootstrapped by V_{ϕ})
A^*	selected action $\arg \max(g + \text{logits} + \sigma(\hat{q}))$
π'	search-improved policy target $\text{softmax}(\text{logits} + \sigma(\text{completedQ}))$
V_{ϕ}, z	leaf-value network; its regression target

PPO improves the policy with a first-order gradient on sampled returns; the AlphaZero family improves it by *planning*. Discretising the action (the categorical WHERE of Section 7) was the prerequisite. Because **GpuEnv** is a perfect, batched, cheaply-cloneable simulator, the **learned model is unnecessary** — AlphaZero (search with the real model), not MuZero, is the template: the encoder + trunk is the representation, the real env step is the dynamics, the real reward is the reward head, the factored actor is the policy prior, and the PPO critic is the leaf value. Three obstacles and their resolutions:

- **Combinatorial factored action** (one categorical per owned planet): sample m joint actions via per-factor **Gumbel-Top- k** (Multiagent Gumbel MuZero) and search only those; cost scales with samples, not the product space. The reachability mask shrinks each row’s support first.
- **Very few simulations: Gumbel MuZero** gives a *guaranteed* policy improvement at any budget — Sequential Halving over the sampled top- m , action $A^* = \arg \max(g + \text{logits} + \sigma(\hat{q}))$, training target $\pi' = \text{softmax}(\text{logits} + \sigma(\text{completedQ}))$. Each q is a depth-1 real env step bootstrapped by V_ϕ .
- **Simultaneous moves / hidden opponent**: treat the frozen league opponent as a *chance node* (opponent-as-chance / Stochastic MuZero afterstates) — we need no learned chance model because μ is a network we can query; averaging over a PFSP mixture makes π' a best response to the population.

The loss replaces the clipped surrogate with $\text{KL}(\pi' \parallel \pi_\theta) + c_V(V_\phi - z)^2$; everything else (factored actor, masks, GPU rollout, league) transfers as-is, and the design degrades gracefully to current PPO at $m=1$. Reanalyze (re-search stored trajectories with the latest network) is a large sample-efficiency win the perfect simulator makes free.